

CSE 460
Spring 26

Quiz-01

Name: _____

Student ID: _____

Scenario 01:

You are given a logic circuit which does parity checking. It means the circuit gives an output 1 whenever it gets an odd number of 1s as input, or else it gives 0 output. The circuit takes 3 inputs at a time.

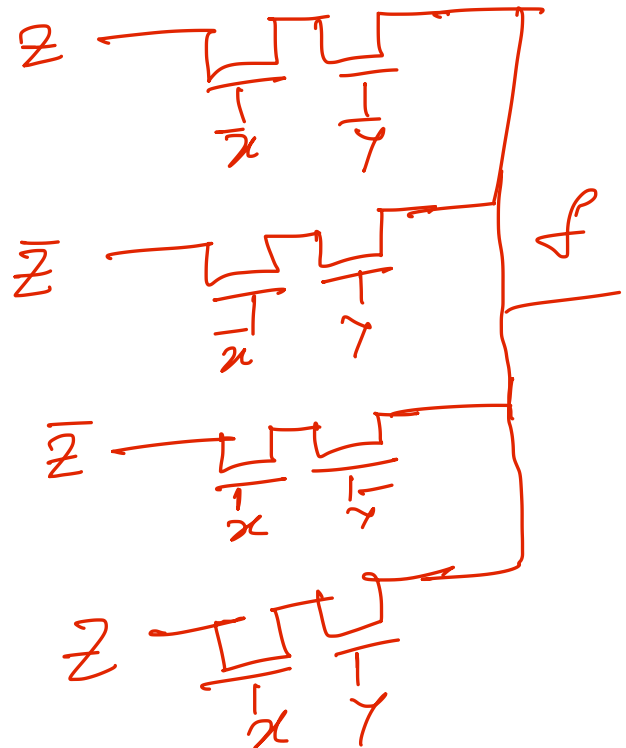
- 1) Identify the logic circuit and draw the truth table.
- 2) Find the logic expression from the truth table and implement it using NMOS pass transistor logic.

→ XOR circuit.

→

x	y	z	f
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

$$f = \bar{x}\bar{y}z + \bar{x}y\bar{z} + x\bar{y}\bar{z} + x\bar{y}z$$



You are given the following circuit-



- neg edge RT

